What's IN a model? Modelling IN services using formal methods

K C Woollard

Over the past twenty years there have been vast changes in the telecommunications business. Most notably there have been moves from analogue to digital transmission and from mechanical to electronic switches with flexible software and protocols allowing a high degree of customisation. These fundamental changes to the infrastructure have enabled the massive change currently going on in the area of advanced service provision. It is now possible to develop and deploy sophisticated services on to the digital network at an ever increasing rate. Unfortunately design and specification methods are still being used that were only suitable for the networks and services of ten years ago. It is essential for the succesful introduction of complex services into a telecommunications environment that they are well specified, documented and understood before being released for service. This paper details activities currently being undertaken to resolve the problems associated with service definition and interworking.

1. Introduction

This paper explores the problems associated with designing and specifying new services and the potential problems of interworking a number of services together. Furthermore it discusses the problem of interworking services across different network platforms, e.g. intelligent network (IN) services interworking seamlessly with ISDN services.

The paper also explores the techniques available to enhance the understanding of a service during the critical stages of design and specification and in particular presents the advantages gained from rigorously modelling the service. The fundamentals of modelling are discussed with particular attention to the key issues of abstraction, focus, simplicity and clarity. The formal methods available for design and modelling are introduced with emphasis on the use of the Specification and Description Language (SDL) [1—3] and its associated tools. Using this technology it is possible to develop a good understanding of the operation of a single service and, when simulated in combination with others, it is possible to study the interaction between multiple services and the basic call control.

The ability to study the interaction between services is highly desirable, especially at the very early stages of service specification [4, 5]. In this way it is possible to highlight the potential interworking problems long before they are introduced into the network where they degrade the quality of service and are costly to rectify.

The paper will detail some examples of problems solved by employing these new techniques and how these are being introduced into the equipment procurement and testing activities.

2. Service specifications and standards

The specification methods employed by BT, and others such as the ITU, have for many years relied on informal techniques such as English descriptions and message sequence diagrams. While these methods have been acceptable in the past, the problems that can arise from such a free format are beginning to be understood. In particular a number of telecomunications operators are reporting interworking problems within their networks.

These problems arise for a number of reasons. The main cause is that new services are developed in isolation from existing services and others under development. For example the ITU ISDN standards were developed in isolation from existing services such as BT's ISDN and centrex. This was partly due to the conflict of national versus international standards. The ITU has developed ISDN in isolation from mobile and IN services. To support these services a number of different, and incompatible, network solutions have been adopted by the telecommunications industry. As a result there are many, and varied, protocols both in the access and core network, and a number of similar but incompatible services. For example, it is possible to identify at least three different call-diversion offerings in BT's network relating to centrex, ISDN and network services. It would suit most network operators and customers if the services and protocols were compatible across all networks.

A further cause of interworking problems is associated with the methods used to define both existing and future systems. Because of the informality of the current methods it is not possible to understand either the precise behaviour of, or the implications associated with, new product definitions.

The need for new services is forcing the pace, with technology expected to deliver faster and faster service provision with a more and more complex set of services. This brings about an interesting paradox where, looking forward, it would be pleasing to see a multitude of new services, highly flexible service creation environments and numerous service providers all seamlessly interworking. However, even with just a few services on the current networks, very little interworking has been achieved.

Because rapid changes are required in these areas, two specific scenarios need to be resolved. For current systems there is the task of identifying problems and solving these by modifying the service in operation. At the same time the requirements of building the vision of tomorrow without making the mistakes of yesterday must be faced. It is clear that there is a need to change working practices by, for example, developing all networks and services in a common context, and technology and methods are needed to help.

3. Seamless services through rapid modelling and proving?

One of the major difficulties that exist in removi ng the current interworking problems is the limited understanding of the detailed operation of the present network and services.

To address this problem BT's service specifiers have started to adopt proven techniques from the product development arena. In particular by applying advanced specification methods, such as SDL, it has been shown that complex products can be developed at lower cost with improved quality. Furthermore, these products can be re-used or partially reused in subsequent developments, and later modifications are easy to realise.

The products that can be developed using formal methods cover most of the software associated with the telecommunications environment, i.e. the switch control software, the protocols and the services. In general these can be considered as separate products, since services in the future will be added with a degree of independence from the underlying network.

The use of SDL in the context of resolving existing network problems is relatively new in BT, but significant progress has been made in investigating services such as network services, ISDN and centrex. It has been shown that an SDL model of the service is able to independently detect known interworking problems such as the fundamental problems between call diversion and call waiting.

These two simple services typify the problem currently faced by BT. Call waiting and call diversion were both developed as network services and have been available on System X switches for some considerable time. It was not until there was sufficient geographical coverage of these switches that BT attempted to launch the services on a national basis. It was then, some time after development, that it was discovered that they did not interwork and could not both be offered to a customer. This is not only a problem for BT as examples abound in other telcos.

The main cause of the shortcoming in product development and testing was the lack of a sufficiently clear and complete specification. This specification has been retrospectively produced as an SDL model and, as a result, the cause of the problem has been identified.

Encouraged by these successes other services have been modelled with similar results. Known interworking problems have been reproduced and more importantly new interworking situations have been identified.

In particular, time has been spent on developing an IN service model as part of a feasibility trial of these methods.

Initially a number of issues associated with the specification of the services were identified. Mostly these were associated with the clarity and completeness of the specification.

Having developed the service definition, the formality of the language and the associated tools can be used to produce a simulation of the system being designed.

The simulation allows the design and behaviour of the services to be studied both in isolation and in combinations with other services. In particular, the service behaviour can be validated against the original requirements, allowing rapid and cheap re-development if it does not conform. Having validated against these requirements the service can be investigated against further sets of requirements such as feature/service interworking and possibly portfolio coherence issues such as the consistent use of tones and announcements. Furthermore, having developed a high quality model, it is an ideal basis for specification, implementation and testing.

The use of SDL to develop service models is leading to the model-based development life cycle shown in Fig 1. The current life cycle still exists in this model with the stages from informal requirements, through specification and development, to the product, and with the parallel path from requirements to test suite. However, Fig 1 shows the addition of the SDL model and its importance at the beginning of the life cycle. In particular there is an iterative process taking place between requirements capture and model development that is essential to generating a clear set of requirements and a complete model. Once the model has been established it can be tested for conformance to the requirements and any inconsistencies can be corrected.

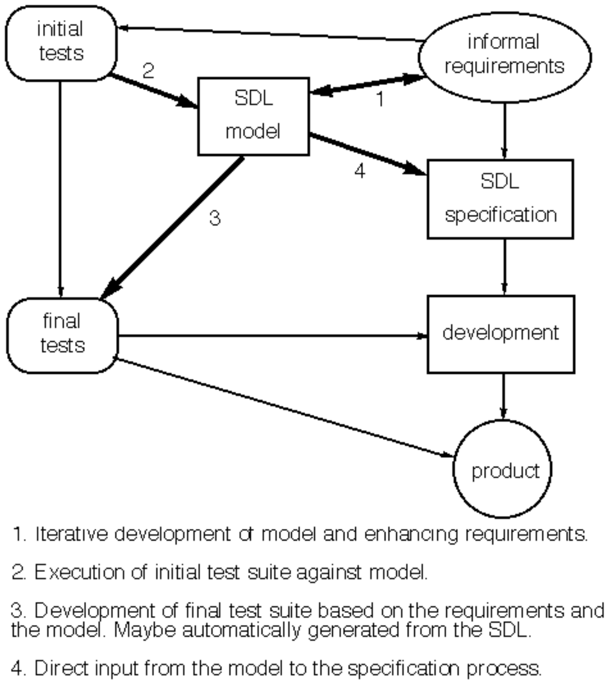

Fig 1 Model-based development life cycle.

Once validated the model can be used in conjunction with the requirements to produce the specification of the product. In this case the use of SDL for the specification is shown and often there is a strong similarity between the specification and the model. Similarly, the model can be used to develop more detailed test cases to enhance those derived from the requirements. These tests can be produced in a semi-automatic way from the SDL model using the available SDL tools.

4. Fundamentals of a good model

The approach being advocated here is a combination between abstract specifications and pure modelling. Indeed both have their place. In the engineering world of complex systems the need for real models has long been accepted. For example, no one would dream of producing a new aeroplane without wind tunnel tests on a model. Similarly all new ships are subjected to extensive model testing in a wave tank.

The reason for extensive modelling in the engineering world is relatively clear. The systems being developed are complex and in some cases there are no sound scientific principles on which to move from specification to implementation with guaranteed success. The examples above are both cases of three-dimensional fluid flow for which there is no complete mathematical model. The opposite may be considered true of bridge building, where the laws of mechanics have proven reliable in the past — in this case the move from engineering drawing to the final product can be made in one stage with a predictable outcome.

In the software world of communications and computing there are very complex systems which are growing in complexity all the time. Like the fluid-flow problem there is science to aid in the development of products. It seems sensible then to resort to the tried and tested principle of building and testing a model before implementing.

Like the model boats, those aspects of a system of interest need to be identified and captured in the models. For example, a toy yacht looks like the real thing, sails like the real thing but is obviously not a real yacht. The toy has the essentials of a hull, keel, mast and sails. It may be constructed in a different material and is usually scaled down in size. It will also have some elements that perform a needed function that are different to the items found on a real boat. All of these issues add up to the concept of abstraction, i.e. representing the real item in a limited but meaningful way. Other issues that need consideration are completeness (does the boat have sails?) and simplicity (are all the features needed clearly visible and easy to use?). For any model to be useful these issues must be satisfactorily treated. One last point about the model boat is that the boat building industry relied for many generations on carefully constructed models for providing all the measurements for the real vessel. This represents the classic example of moving from requirements to model to implementation.

Developing specifications at the correct level of abstraction is essential to successful specifications. Too much information and implementation choice is restrained, too little and the specification is incomplete, leading to divergent products.

In the IN the services are separate from the underlying network and its protocols. As a result the abstract notion of a basic call is more important for the IN than previous systems. Having accepted the need to include a basic call within the service specification, an abstract model that is independent of hardware or protocols can be defined. The role of this basic call is to represent the behaviour of the network that would otherwise be missing from a purely service level definition.

In this way the basic call is not intended for direct implementation, as is the case for the services, but rather it should have a traceable relationship to the more detailed network and protocol specifications associated with an implementation. The relationship between the basic call and its service models and the real network is shown in Fig 2.

This mix of functional separation and abstraction within the specification provides the ability to focus clearly on the issues that need to be resolved in the model. In the definition of IN services the issues pertain to the customer-to-service relationship, service-to-network (basic call) and service-to-service. Each of these issues must be covered within the service specifications and each area can be refined into a more detailed set of issues.

- Customer-to-service relationship

  In the relationship between the customer and the service the concern is that the presentation of the service is consistent with other service offerings. This is usually termed the 'look and feel' of a service and covers such things as the service invocation sequence (e.g. * # codes), the recorded announcements played and the way the service completes. f, these elements of the specifications are not controlled, it is unlikely that the customer will experience coherence in ease of use.

- Service-to-network relationship

  The area most covered in IN service specifications is the relationship between the services and the basic call. This is not very surprising as this represents the 'usual' operation of the network. Of course, this is an over-simplification but, in general, it is essential to understand how the service is provided by the network.

- Service interworking

  The third area identified as of interest is that of service interworking. This is currently a very active research area due primarily to the complex nature of the problem. In general, the services implemented are relatively few and simple, with limited interactions. However, telecommunications networks world-wide have started to experience interworking problems between services. As described earlier, even simple services have been found to be incompatible, requiring network upgrades. It is clear that if problems exist with current services, many more problems will be seen in a service-rich IN. The first line of defence is to improve the specification techniques by considering, modelling and hence managing the interworking of services.

For any specification a clear set of functional requirements is needed and the act of modelling helps to achieve these. For IN services specifications of the service behaviour, interworking with the basic call, interworking with the customer and interworking with other services are needed.

By imposing the above considerations on model building, an attempt is being made to produce specifications that are focused on the important issues, are as simple as possible and hence as clear as possible, and, lastly, as complete as necessary to fully define the service.

5. Methods of modelling

Most of the discussion above is generic and can be resolved by one of a number of specific methods.

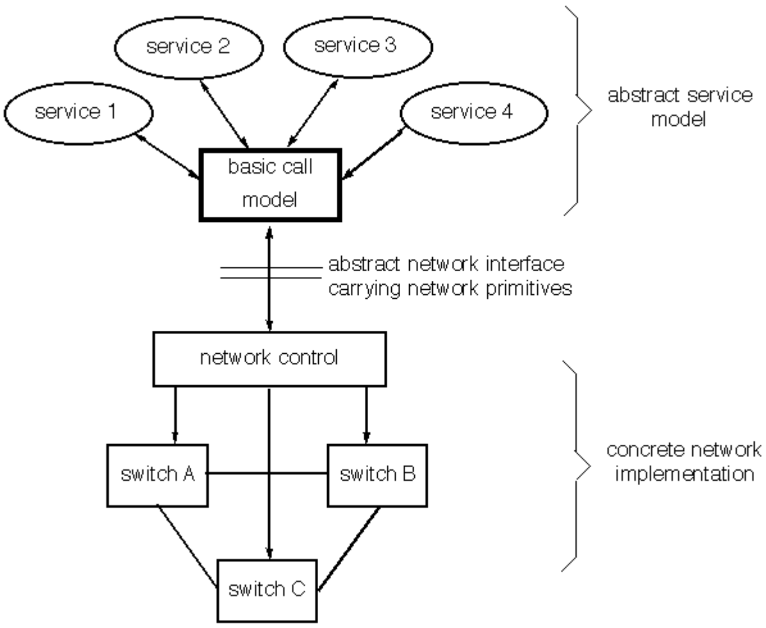

Fig 2 Relationship between service models and network elements.

It is possible for the specification and modelling activities to be separated, e.g. with specifications written in English and rapid prototypes developed in the software language ‘C’. The problem with this approach is the high level of effort required to produce both a good English specification and a ‘C’ prototype. Furthermore, there is no semantic relationship [1] between the specification and the prototype. In fact this gives an abstract specification and an example implementation. The lack of a semantic link means that other implementations may still be demonstrated to conform to the specification yet exhibit behaviour incompatible with the prototype.

A more ideal solution would be one with a very strong link between the specification and the prototype model. To provide such a solution the technology of formal languages has been developed. These formal languages strive to provide the best of both worlds, i.e. the expressive power of informal languages is needed to allow the descripton of things in an implementation-independent and abstract way. At the same time a strong semantic basis is required so that tool support can be developed and the designer can reason about the behaviour of the system based on the specification. The strong semantic basis allows the specification to maintain a strong link to both prototypes and implementations.

The use of such an environment at last allows the addressing of the outstanding problems of incomplete and imprecise specifications.

Examples of formal languages include SDL, ESTELLE, LOTOS from the standards world, Z and VDM from the software world and a whole host of specialist languages from academia such as CCS and CSP.

Each of these notations has good and bad points and can be applied in different circumstances to different effect. The skill is in selecting the appropriate method for the requirements.

In the telecommunications industry there is a growing support for the use of SDL as a method for both specifying and implementing systems. There are a number of reasons why SDL is gaining wide acceptance:

- it has an easy to use graphical syntax,
- it is based on finite state machine theory, which is highly suited to distributed message-passing systems such as a telecommunications network,
- it has excellent tool support that allows full system analysis and validation of the specification,
- it is possible to automatically produce prototype models from the specification and, in a development environment, real code can be produced at the press of a button.

It is possible that in the future the use of formal specifications will be seen on the interface between BT and its suppliers. This could have advantages both at the procurement stage and during testing and integration.

The use of SDL as the specification language produces clear and precise definitions that, with care, can support the appropriate level of abstraction. If needed, specifications can be developed that contain a significant level of detail and hence limit the number of options for the implementor.

There is a general problem of abstraction associated with all specification, modelling and implementation work. Certainly it is true that standards bodies such as ITU do not wish to produce specifications biased towards a particular implementation and hence manufacturer. At the same time specifications should not be so open that there is little chance of interworking. Currently there appears to be no panacea to this problem and each exercise relies on the judgement of technical experts.

Whatever the conclusion, the use of SDL should help the development process. In the case where the specifications are very detailed it is possible to consider code generation directly from the specification. This is an option being taken up by many of the product developers in the telecommunications industry with encouraging results.

If the specifications are very abstract, then the ability to simulate them allows the opportunity to clearly understand the behaviour required. It is this ability to model in an abstract plane that is central to the work on IN service modelling. The prime aims of this modelling work are to validate the service specifications and to detect any unwanted system behaviours, such as feature interworking problems.

6. Modelling IN services

A number of activities have been carried out within BT to assess the usefulness of SDL for specification, modelling, and implementation. The findings for implementation are clear and are detailed in Cookson and Woodsford [6] and are echoed by the findings of others in the industry.

In all the specification work undertaken it has been found that the act of producing a formal specification has posed a lot of questions about the original set of requirements, and led, in general, to moving from an

[1] The semantics of a language gives the precise meanings of the elements of the language. It is semantics that allows the extraction of the meaning from statements in the language. Formal languages such as SDL have a good semantic definition based on mathematics and it is possible to build tools to implement those semantics. A semantic relationship implies that there is a provable and reproducible relationship between two representations of the same thing.

informal set of specifications in English and Message Sequence Charts, to a formal specification in SDL. The formal specification is invariably more complete and of a higher quality. This means there are less opportunities for implementors to produce the wrong product, the development time should reduce and there should be fewer interworking problems.

SDL has been applied to the specification of a number of BT's proposed services with a number of interesting results. Detailed service design documents were accepted as input to the process. From these documents an independent set of SDL specifications was produced. The SDL specification demonstrated a high degree of correlation to the detailed designs indicating that the informal documents were of good quality.

Once the formal model was complete it was primarily used to look for unwanted feature interactions within the services. This was a manually driven simulation of various service combinations. In this way, it was possible to investigate the user's perspective of the individual services, the interworking of the services as a single package and the interworking between new services and existing network services.

The investigations highlighted 14 potential issues that may affect service interworking (see Table 1). All of these issues related directly to deficiencies in the specifications. Some would most probably be picked up during product development and should be reflected back into subsequent issues of the specifications. Others arise from areas outside the scope of the existing specifications. In particular, a number of problems identified were in the interworking between the new service and existing network services not considered in the service design.

These independent services are seen as the most problematic as they are likely to be developed by different product development teams and launched on to a platform with limited thought about their interworking issues. Furthermore, as the number of services and the service complexity increases, it will become more difficult to cover all the interworking issues in an informal way. Consequently, the results in this area are of particular interest as the modelling work offers a method of studying interworking issues at the specification level.

It is beginning to become accepted that service modelling will be a necessary requirement for all services before launch. It is essential that service modelling is carried out as early as possible in the development process.

Table 1 Service modelling sample results.

	Modelled Service			
Modelled Network Feature	Chargecard	Personal Numbering	Remote Access to Call Divert	Remote Access to Call Minder
Calling number delivery	Fail	Pass	N/A	N/A
Call return	Fail	Fail	N/A	N/A
Call waiting	Fail	Fail	N/A	N/A
Three-way calling	Pass	Fail	N/A	N/A
Call divert immediate	Pass	Fail	Pass	N/A
Call divert on 'busy'	Pass	Fail	N/A	N/A
Call divert on 'no reply'	Pass	Fail	N/A	N/A
Call back when free	Pass	Fail	N/A	N/A
Number translation	Pass	Pass	N/A	N/A
Callminder	Fail	Fail	N/A	Pass
Chargecard	N/A	Pass	N/A	N/A
Personal numbering	Pass	N/A	N/A	N/A
Service access	Pass	Pass	Pass	Pass

Pass: indicates no interworking problems found.
Fail: indicates potential problems found.
N/A: means no tests are applicable.

The findings from the studies and the application of the method on a wider range of products are currently being assessed. One of the areas that is already affected by the findings is product testing [7, 8].

Currently BT tests new products on an end-to-end service basis, making the assumption that satisfactory conformance testing was performed by the manufacturers. During the testing of new products it is essential to uncover as many problems as possible and the issue of service interworking is becoming paramount. Consequently, the results from service modelling are being used as the basis for testing the real product. There are two good reasons for such an approach:

- firstly, the results from the model are focused on the areas of the specification that are weakest and hence most prone to arbitrary decisions during manufacture,

- secondly, the interworking examples are too complex to be defined and results to be predicted in the usual (manual) way.

In these complex areas the services can be investigated as models and the results captured as message sequence charts. These charts can then be used to guide the testing and act as the expected test results.

7. International perspective

There is currently a lot of activity within other companies and international fora concerned with the application of languages, such as SDL, to specification and modelling. In particular there is a clear focus on the new technology areas such as IN and accepted problem areas such as feature and service interworking.

There are activities within Eurescom (project P230 and proposal PP 5302/5) using SDL to address feature interworking in the intelligent network. Similarly the ACTS programme (the European initiative on advanced communications technology and services) is proposing work in the area of modelling and feature interworking.

Some suppliers can be seen to be making a strong move towards the use of SDL as a natural part of the product development life cycle. For the product developers the long promised ‘holy grail’ of automatic code generation is becoming a reality with informally quoted improvements in both coding efficiency and the number of errors produced in the code. Accepted figures for lines of delivered code vary between 100 and 400 per man month using conventional methods. From projects within BT, figures nearer to 5000 delivered lines per man month can be seen using SDL as part of the development. This is an order of magnitude improvement.

Furthermore, the use of SDL as the design language has improved the link between designs and implementations with the documentation being maintained as products are developed and enhanced. Manufacturers are also beginning to identify project management improvements from the use of SDL as the product can be split up, developed and then integrated in a controlled way.

There is also evidence that other telcos, in the UK and elsewhere, are pressing to use SDL on the interface with their suppliers.

Most of the advantages apparent to our suppliers and competitors are consistent with the findings from BT’s work. It is becoming more evident that formal specification, modelling and implementation can significantly contribute to the quality and timeliness of product developments. Furthermore, the power associated with formal method tools can be used to help contain some of the complex problems such as feature interworking.

8. Conclusions

The use of formal models has been shown to offer advantages in a number of areas.

For the first time there is the opportunity to document the capabilities of the current networks allowing new products, or services, to be assessed against existing products. In this way BT has direct control of the product portfolio coherence across the services on offer.

Also, this provides a much enhanced understanding of requirements from manufacturers, reducing the ‘confusion resolution time’ currently associated with equipment contracts.

The opportunity exists both to rigorously assess the delivered product against BT’s requirements and to maintain a tight control on the quality of network services.

It is true that the methods described in this paper will add extra effort to the front end of the development life cycle, but benefit can be gained from all the advantages with the added bonus that overall product development times should be reduced.

References

1 ITU-T recommendation Z.100 and annexes A, B, C, E, F1, F2 and F3.

2 Saracco R, Smith J R W and Reed R: ‘Telecommunications systems engineering using SDL’, North-Holland (1989).

3 Belina F, Hogrefe D and Sarma A: ‘SDL: with applications from protocol specification’, Prentice Hall (1991).

4 Bryce K, Crowther M and King J: ‘Feature interworking detection using SDL models’, Globecom 94, San Fransisco (1994).

5 Crowther M: ‘Modelling and validation of telephone network signalling’, SDL’93, Using Objects, Darmstadt (1993).

6 Cookson M and Woodsford S: ‘Design methodology using SDL’, BT Technol J, 11, No 4 , pp 16—24 (October 1993).

7 Kelly B, Webster N and Bruce G L: ‘Future testing methods using SDL’, BT Technol J, 11, No 4, pp 25—34 (October 1993).

8 Kelly B, Webster N, Boullier L, Phallipou M and Rouger A: ‘Evaluation of some test generation tools on a real protocol example’, 7th International Workshop on Protocol Test Systems, Japan (November 1994).

Kevin Woollard joined BT in 1975 and worked as a technician until he graduated in 1983 with a 1st Class Honours degree in Physics with Physical Electronics from the University of Bath.

He has worked on the specification implementation and testing of ISDN signalling standards and was part of the team to produce the first demonstration of ISDN services in this country. This work led directly to involvement in producing European ISDN test suites in the standard test notation TTCN and the application of the formal description technique SDL to protocol specifications.

He currently leads a group investigating the use of formal notations for specifying BT’s signalling protocols and their application in automatic verification and test generation. This work involves a number of projects including the collaborative European research project PROVE.